

Solution: $u^T = [-2 \ 3 \ -4]$ (By defn. of transpose)

$$u^T v = [-2 \ 3 \ -4] \begin{bmatrix} a \\ b \\ c \end{bmatrix} = -2a + 3b - 4c \text{ (By Row-Vector Rule)}$$

$$v^T = [a \ b \ c] \text{ (By defn. of transpose)}$$

$$v^T u = [a \ b \ c] \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} = -2a + 3b - 4c \text{ (By Row-Vector Rule)}$$

$$u v^T = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} [a \ b \ c] = \begin{bmatrix} u_a & u_b & u_c \end{bmatrix} \text{ (By defn. of matrix multiplication)}$$

$$u_a = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} a = \begin{bmatrix} -2a \\ 3a \\ -4a \end{bmatrix} \text{ (Scalar multiplication)}$$

$$u_b = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} b = \begin{bmatrix} -2b \\ 3b \\ -4b \end{bmatrix} \text{ (Scalar multiplication)}$$

$$u_c = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} c = \begin{bmatrix} -2c \\ 3c \\ -4c \end{bmatrix} \text{ (Scalar multiplication)}$$

$$\therefore u v^T = \begin{bmatrix} -2a & -2b & -2c \\ 3a & 3b & 3c \\ -4a & -4b & -4c \end{bmatrix}$$

$$v u^T = \begin{bmatrix} a \\ b \\ c \end{bmatrix} [-2 \ 3 \ -4] = \begin{bmatrix} v(-2) & v(3) & v(-4) \end{bmatrix} \text{ (By defn. of matrix multiplication)}$$

$$v(-2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} (-2) = \begin{bmatrix} -2a \\ -2b \\ -2c \end{bmatrix} \text{ (Scalar multiplication)}$$

$$v(3) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} (3) = \begin{bmatrix} 3a \\ 3b \\ 3c \end{bmatrix} \text{ (Scalar multiplication)}$$

$$v(-4) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} (-4) = \begin{bmatrix} -4a \\ -4b \\ -4c \end{bmatrix} \text{ (Scalar multiplication)}$$

$$v u^T = \begin{bmatrix} -2a & 3a & -4a \\ -2b & 3b & -4b \\ -2c & 3c & -4c \end{bmatrix}$$